

Bayesian Calibration of a Surrogate Cahn-Hilliard Model to Molecular Dynamics Data.

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Background

- Many physical systems we care about have time and resource barriers that prevent wide optimal design tests.
- We want a faster and robust predictive surrogate model.
- A physical system can be represented through partial differential equations, serving as a physics-informed surrogate model.
- Bayesian Calibration Framework.**
- The surrogate model parameters are unknown, and their relationship to the system must be discovered.
- The predictive capabilities of these models depends on accurately inferring the unknown parameters while quantifying uncertainty.
- Self-assembling nanoscale block copolymers (BCPs):** Important in nanolithography for semiconductor manufacturing, nanoporous membranes, drug delivery, more.
- Current test methods such as Molecular Dynamics (MD) simulations offer high-fidelity data as alternative to laboratory data, but expensive and inefficient to run.
- A surrogate phase-field model based on the Cahn-Hilliard free energy functional offers a more efficient approximation to BCP systems.
- Goal:** Develop a Bayesian calibration framework that efficiently identifies plausible regions of Cahn-Hilliard parameter space and quantifies uncertainty based on observed Molecular Dynamics simulation data.

$$E[c] = \int_{\Omega} \left[\underbrace{\frac{1}{4}(c^2 - 1)^2}_{\text{phase separation}} + \underbrace{\frac{\epsilon^2}{2} |\nabla c|^2}_{\text{interface penalty}} \right] dV + \underbrace{\frac{\sigma}{2} \int_{\Omega} (c - m)(-\nabla^2)^{-1}(c - m) dV}_{\text{long-range interaction}}$$

Figure 1: Cahn-Hilliard free energy functional with a non-local Ohta-Kawasaki term for long range interactions. A spatiotemporal model of BCP systems.

Abstract

This project aims to develop a Bayesian calibration framework for a surrogate model for a faster and accurate predictive alternative to Molecular Dynamics simulations which are slow and inefficient. Three parameters govern the Cahn-Hilliard model, $\Theta = \{\epsilon, m, \gamma\}$. Variation of Θ results in different surrogate models. We want to solve this system identification problem by finding the surrogate models that best fit the 5 high-fidelity Molecular Dynamics simulations we have to calibrate against. Bayesian inference allows us to quantify uncertainty about parameter regions by estimating a posterior distribution over parameter space. We implement a solver in C++ and run on a high-performance computing (HPC) cluster. 3D fields generated by both MD and the surrogate model are mapped to a 2D power spectrum, from which 3 scalar Quantities of Interest (QoIs) can provide information about the concentration field. A Gaussian Process machine learning model (an emulator) is trained on an ensemble of Cahn-Hilliard simulations, which is more efficient than solving the Cahn-Hilliard partial differential equation and quantities of interest. This allows efficient Markov Chain Monte Carlo sampling of the posterior distribution using a No-U-Turn (NUTS) sampling algorithm.

Cahn-Hilliard Surrogate Phase-Field Model

$$\frac{\partial c}{\partial t} = \nabla^2 \left(\frac{\delta E}{\delta c} \right) = \nabla^2 (c^3 - c - \epsilon^2 \nabla^2 c) - \frac{1}{\epsilon^2 \gamma^2} (c - m).$$

Figure 2: Evolution equation for the Cahn Hilliard phase field model.

The chemical potential is derived by taking the variational derivative of the free energy functional (Figure 1), evolving the concentration field towards lower free energy over time. The resulting partial differential equation (Figure 2) is solved using an implicit backwards differentiation formula (BDF1) solver implemented in C++.

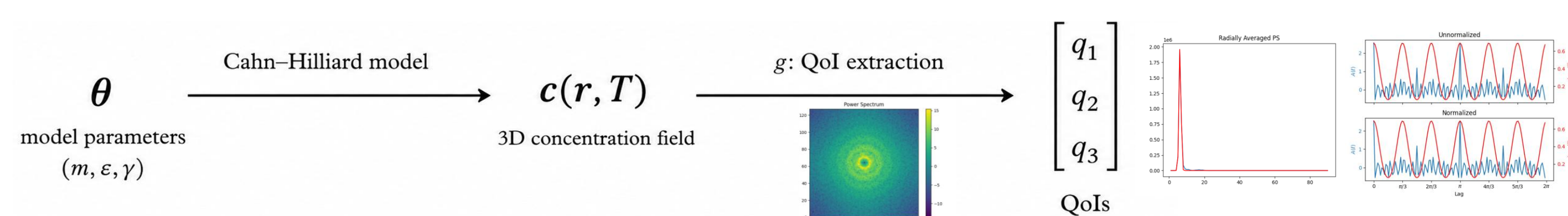


Figure 3: Quantities of Interest (QoIs) are obtained by mapping 3D concentration field to 2D power spectrum, to 3 scalar QoIs.

How can we compare differently parametrized simulations that aim to model the same BCP system? Due to the periodic nature of the final structures, 3 Fourier-based scalar Quantities of Interest (QoIs) are extracted from the 3D concentration fields produced by Molecular Dynamics and Cahn-Hilliard models.

Bayesian Calibration Methods.

Why Bayesian Calibration?

- The Cahn-Hilliard partial differential equation is a continuum approximation of the block co-polymer system.
- Multiple parameter sets may produce similar morphologies (parameter non-identifiability).
- We cannot confidently estimate a single best-fit parameter set to calibrate the Cahn-Hilliard model due to limited observations and parameter non-identifiability.
- We want to Infer a posterior distribution over Cahn-Hilliard parameter space consistent with Molecular Dynamics data.
- Posterior samples quantify uncertainty and parameter correlations in the model.

Markov Chain Monte Carlo Methods (MCMC)

- MCMC allows us to generate representative samples from the posterior distribution.
- No-U-Turn Sampler (NUTS) algorithm to sample parameter space.

Gaussian Process machine learning emulator (Figure 4):

- Calibration becomes inefficient if the Cahn-Hilliard PDE must be solved thousands of times for MCMC sampling.
- A Gaussian Process machine learning model is trained on an ensemble generated by the Cahn-Hilliard surrogate model to learn the mapping between Cahn-Hilliard model parameters to corresponding quantities of interest.

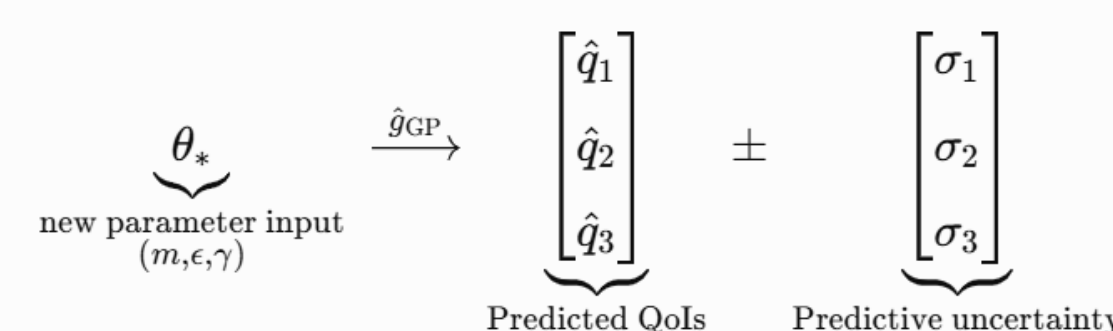


Figure 4: Gaussian Process emulator learns the mapping of the surrogate model parameters to quantities of interest, and uncertainty measurements.

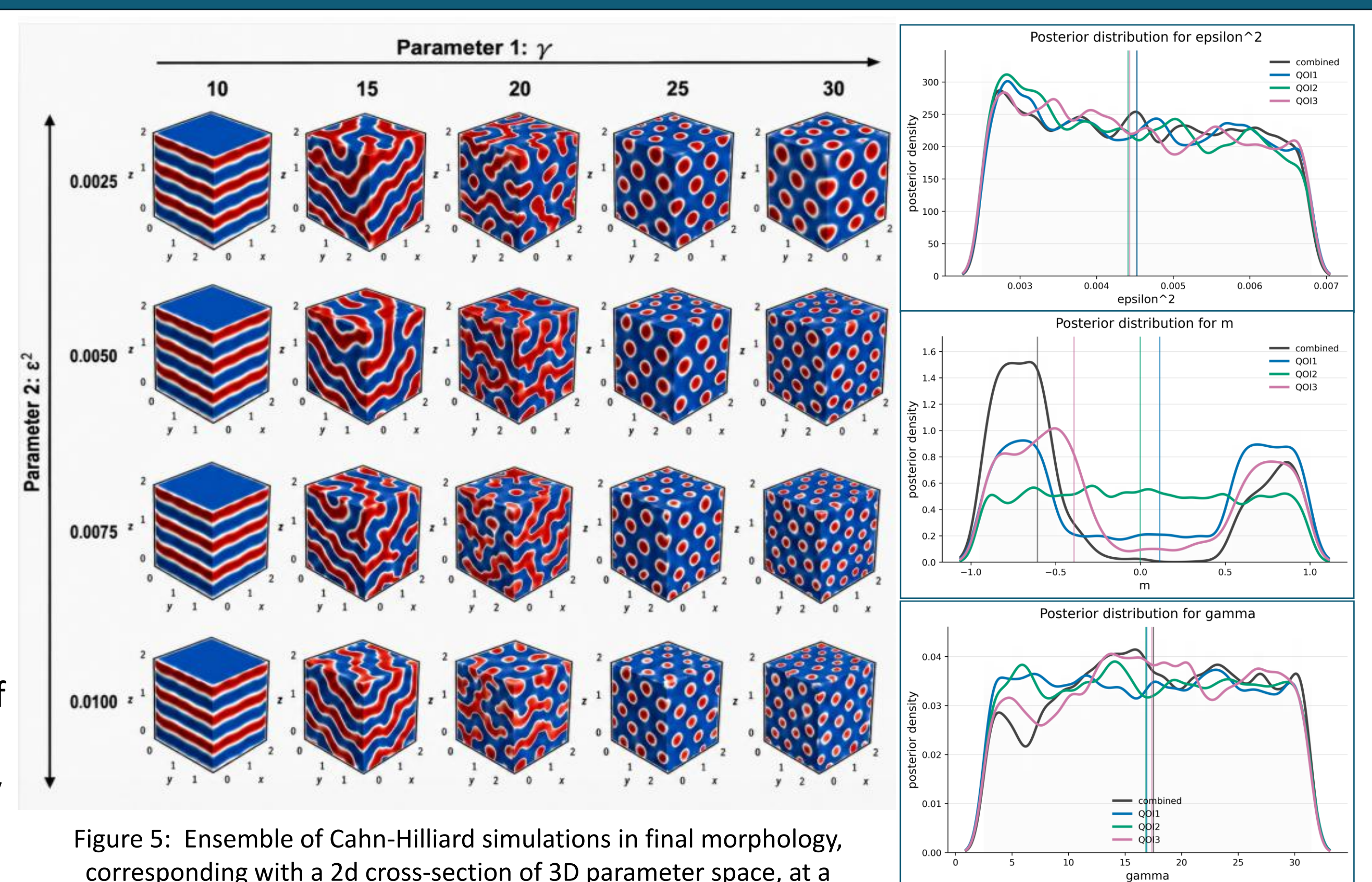


Figure 5: Ensemble of Cahn-Hilliard simulations in final morphology, corresponding with a 2d cross-section of 3D parameter space, at a boundary where morphologies transition. Observe stable stripe, dot, and metastable lamellae structures between.

Figures 6, 7, 8: Posterior distributions of the 3 parameters $\Theta = \{\epsilon, m, \gamma\}$ in the Cahn-Hilliard PDE, conditioned on each of the QoIs separately, and on all 3 combined.

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Conclusion

- Conditioning on the 3 Quantities of Interest (QoIs) produced consistent posterior distributions, indicating agreement among these observables over plausible Cahn-Hilliard parameter values.
- QoI 2 provided comparatively weaker parameter discrimination than QoI 1 and 3. QoIs that are more sensitive to changes in model parameters are more informative in Bayesian calibration because they more effectively distinguish among competing parameter values during system identification.
- The m posterior is largely multimodal. The posterior for γ remains nearly uniform across its physical range, revealing practical non-identifiability. The ϵ^2 posteriors are mutually consistent but broad.
- This Bayesian Calibration framework exposes plausible parameter regions allowing uncertainty-aware system identification.
- It a systematic approach for quantifying parameter uncertainty and calibrating surrogate models, establishing a foundation for future surrogate-based materials design.

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